DO NOW — RECALL

Lesson 140 · Referential Integrity, Indexing and Keys

Name: _____ Class: _____ Date: _____

1. What is the result of UPDATE without a WHERE clause?

2. What is a trace table used for?

3. How many bits are in one byte?

4. What data structure uses LIFO?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Every row in the target table can be updated.

2. Model answer

Tracking variable state changes as an algorithm runs.

3. Model answer

Eight.

4. Model answer

Stack.

By the end, you can…

• Order.CustomerID references Customer.CustomerID.

• Invalid FK values are rejected unless permitted rules apply.

• An index can speed lookup at the cost of extra storage/write overhead.

Vocabulary — know these terms

referential integrity — Foreign-key values correspond to existing referenced keys or permitted NULLs.

index — Auxiliary structure helping the DBMS find rows efficiently.

primary key — Unique identifier of each record.

foreign key — Attribute referencing another relation.

NEW LEARNING

Read, annotate and explain

Link the records

Order.CustomerID references Customer.CustomerID.

Enforce integrity

Invalid FK values are rejected unless permitted rules apply.

Improve searches

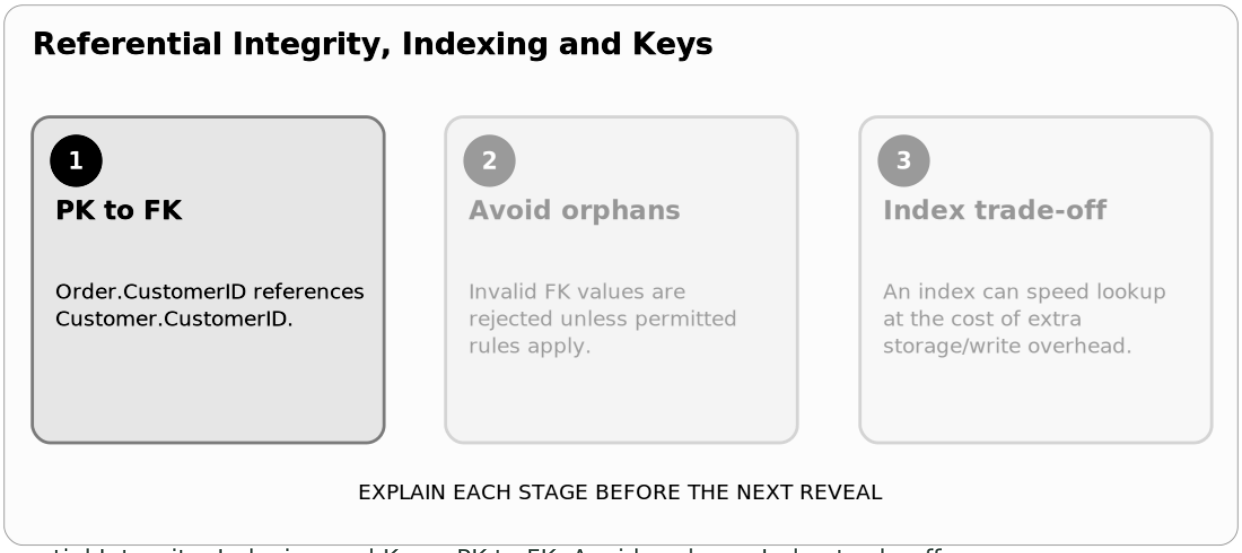
An index can speed lookup at the cost of extra storage/write overhead.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

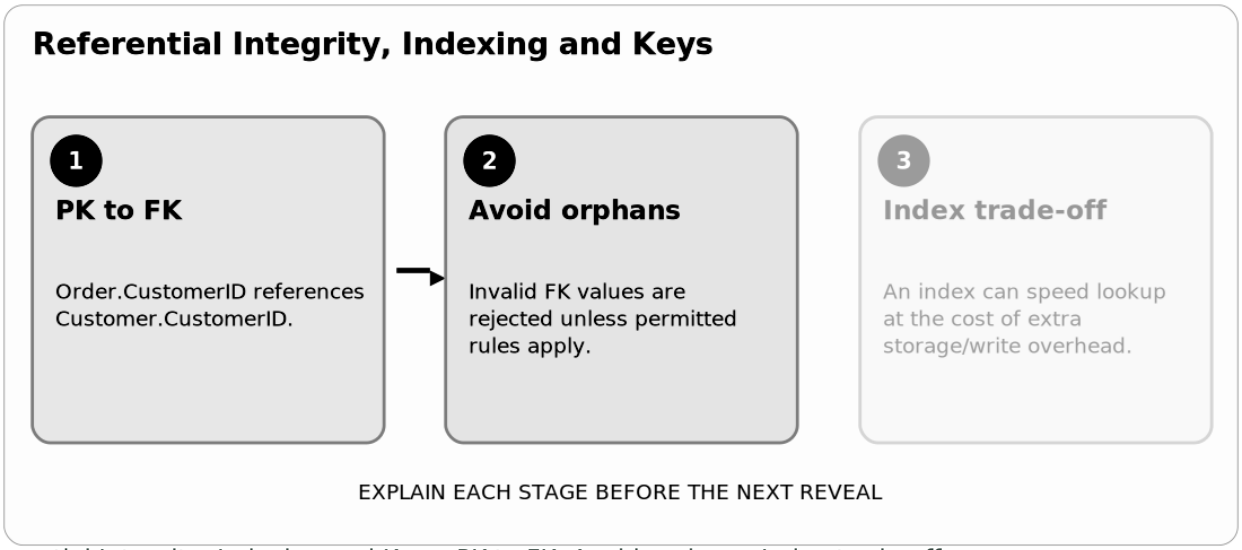
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Referential Integrity, Indexing and Keys: PK to FK, Avoid orphans, Index trade-off

New Learning

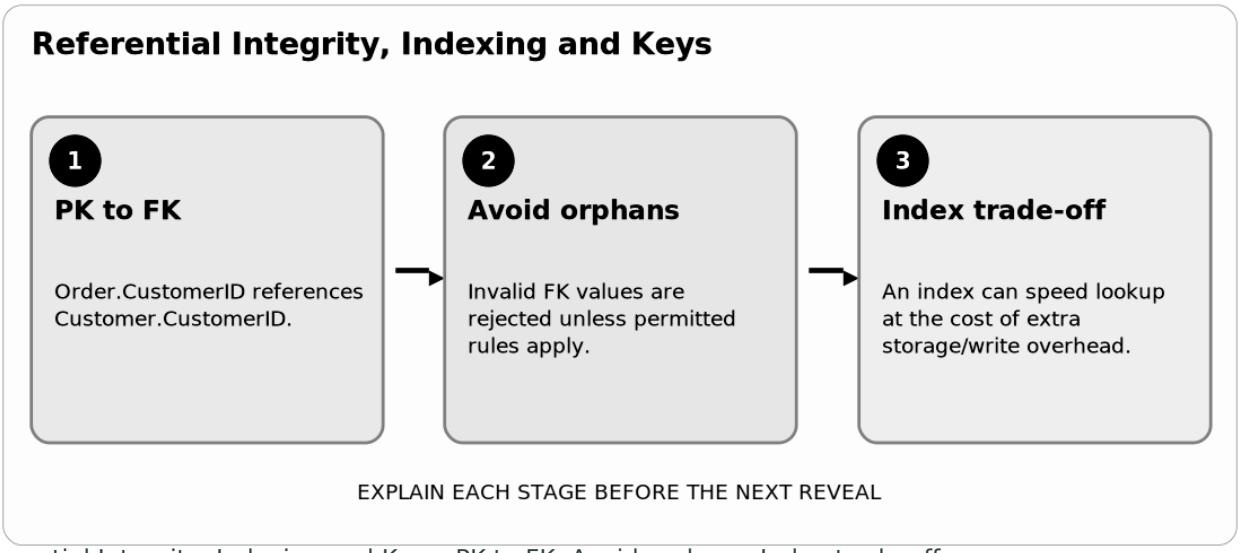


Referential Integrity, Indexing and Keys: PK to FK, Avoid orphans, Index trade-off

New Learning · original lesson diagrams

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New Learning



Referential Integrity, Indexing and Keys: PK to FK, Avoid orphans, Index trade-off

I DO — WORKED EXAMPLE

Follow the worked method

Customer has IDs 7 and 9. A new Order has CustomerID 12. Explain what happens under enforced referential integrity.

Customers 7 and 9

Identify the relevant information and make a prediction.

Order requests ID 12

Explain the change using correct technical terms.

Reject orphan reference

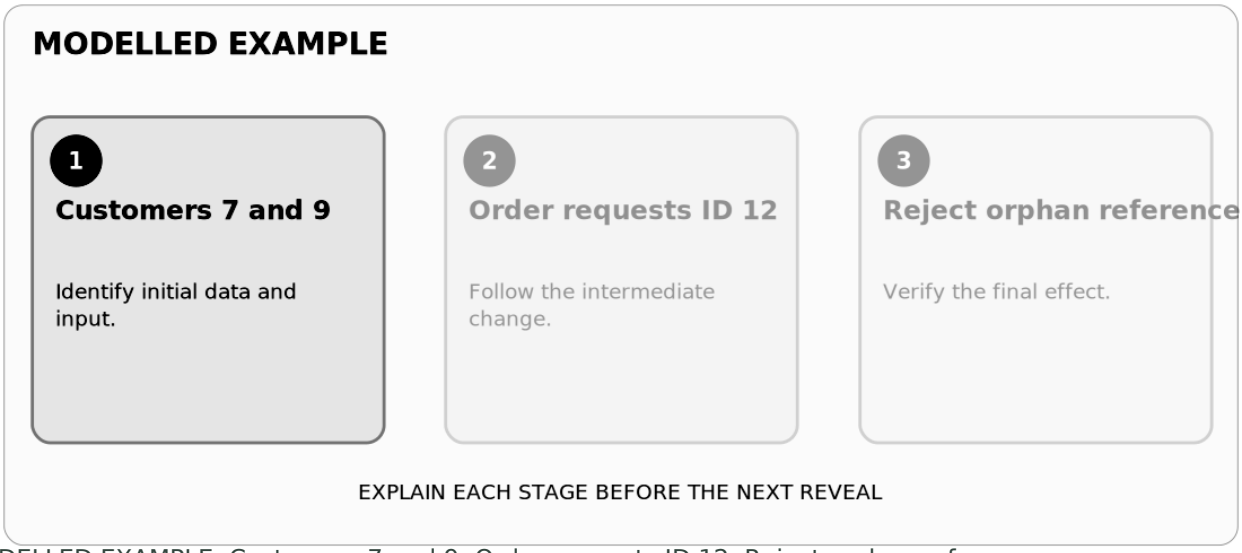
The DBMS rejects the foreign key value 12, unless a matching Customer row is created or the reference is changed. Policies for deleting parent rows may RESTRICT, CASCADE or SET NULL, where permitted and configured.

I Do · original lesson diagrams

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Customer has IDs 7 and 9. A new Order has CustomerID 12. Explain what happens under enforced referential integ

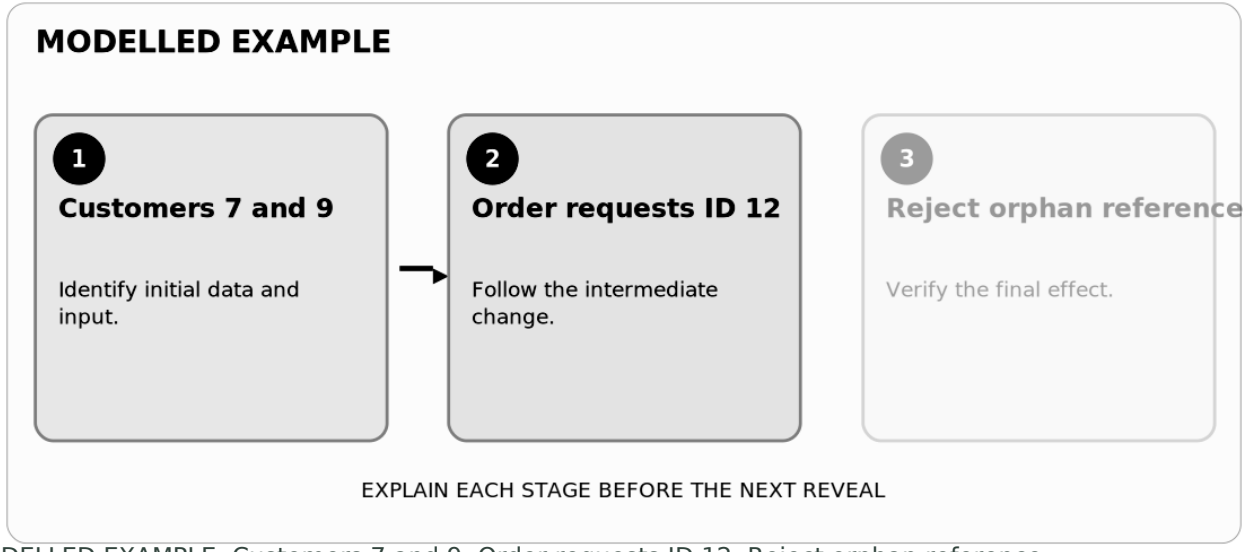
Identify the relevant information and make a prediction.



MODELLED EXAMPLE: Customers 7 and 9, Order requests ID 12, Reject orphan reference

Customer has IDs 7 and 9. A new Order has CustomerID 12. Explain what happens under enforced referential integ

Identify the relevant information and make a prediction.



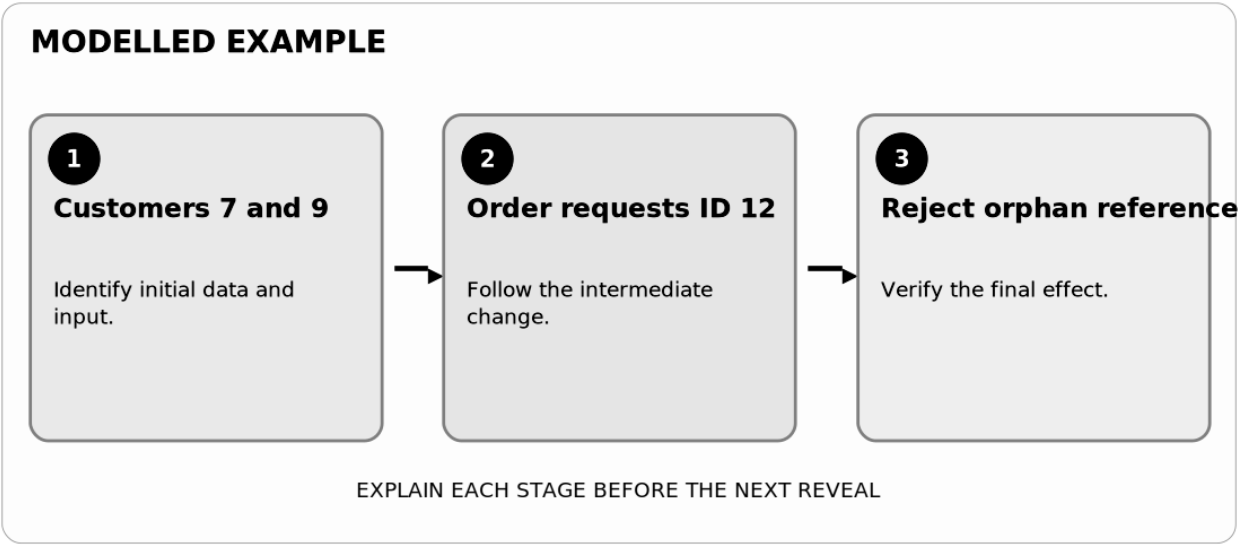
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I Do · original lesson diagrams

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WE DO — GUIDED PRACTICE

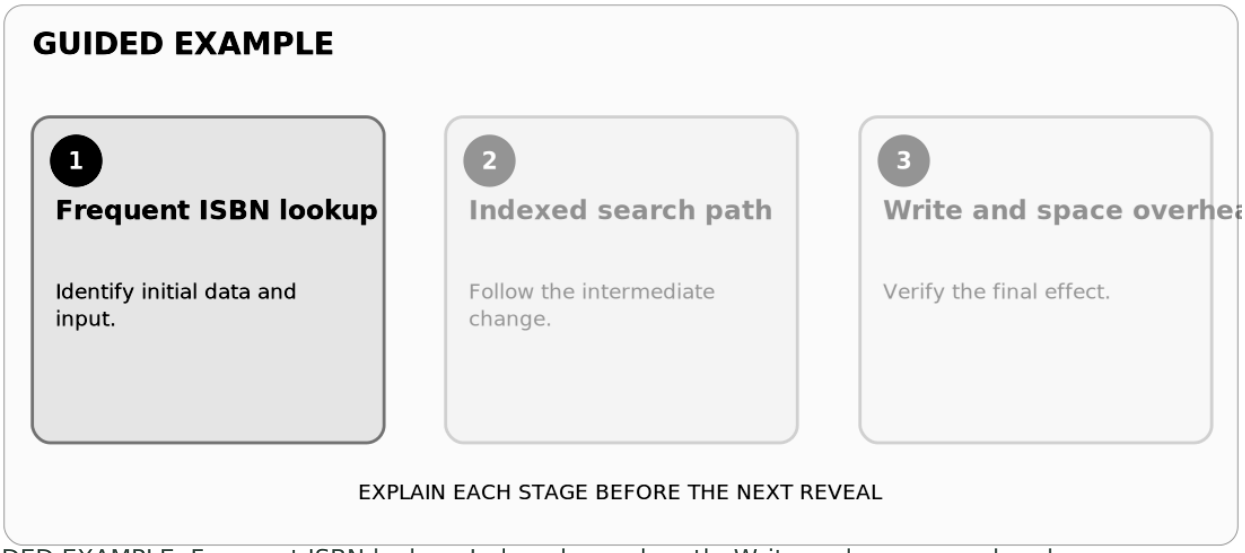
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A large Book table is queried by ISBN many times per day. Should the ISBN have an index?

Identify the relevant information and make a prediction.



GUIDED EXAMPLE: Frequent ISBN lookup, Indexed search path, Write and space overhead

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Frequent ISBN lookup Identify the relevant information and make a prediction.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain referential integrity using Borrower and Loan with a suitable example. [6]

Question 2

Identify two benefits and one cost of indexing a frequently searched field. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A library deletes Borrower ID 8 while Loan rows refer to 8. Compare RESTRICT and CASCADE and justify a safe policy. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An index guarantees every column value it indexes is unique.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

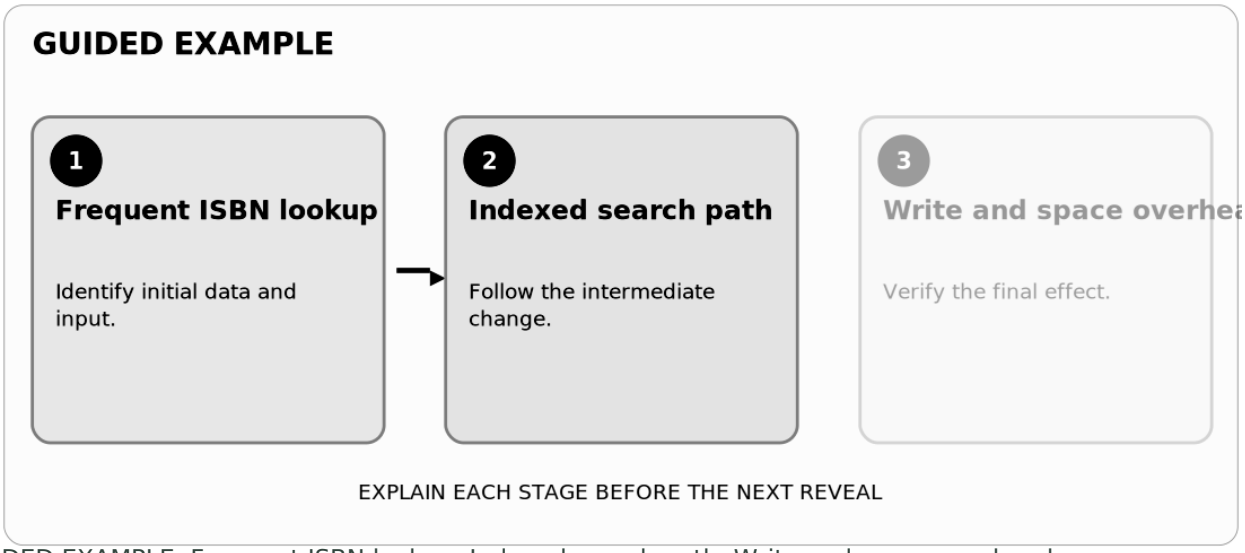
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A large Book table is queried by ISBN many times per day. Should the ISBN have an index?

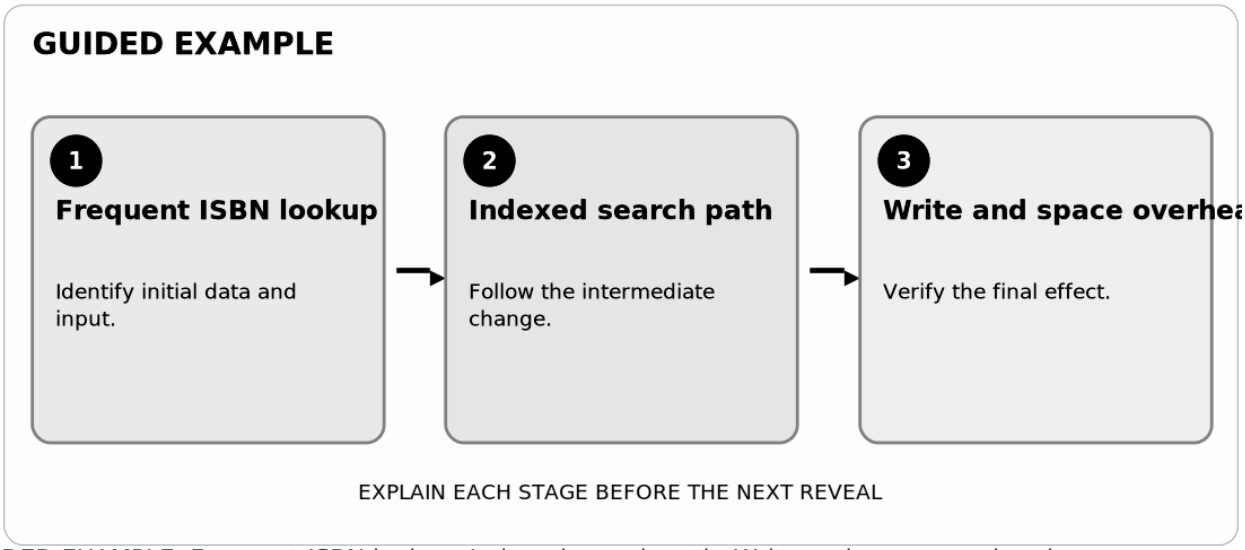
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GUIDED EXAMPLE: Frequent ISBN lookup, Indexed search path, Write and space overhead

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Identify the relevant information and make a prediction.



GUIDED EXAMPLE: Frequent ISBN lookup, Indexed search path, Write and space overhead

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Indexed search path

Explain the change using correct technical terms.

Write and space overhead

An index on ISBN can support quicker exact-match retrieval, especially for large data, but occupies space and requires maintenance after insert/update/delete. A UNIQUE constraint is appropriate if ISBN must be unique in this table.

IMPROVE: Explain one correction to your own reasoning.